

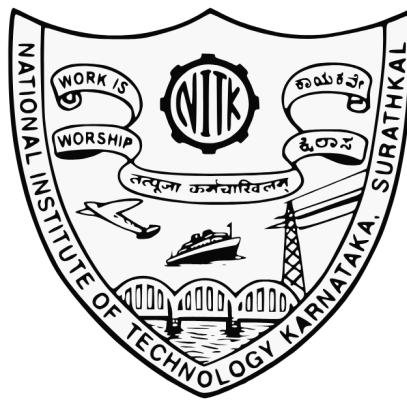
CS754 - MAJOR PROJECT

Progress Report

Study of Double Roman Domination in Unit Disk Graphs

Under the guidance of

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1 Introduction

Domination parameters on geometric graphs (notably UDGs) capture many practical problems in wireless networks, facility location, and resource allocation. The *double Roman domination* (DRD) number $d_R(G)$ extends Roman and classical domination by allowing values from 0, 1, 2, 3 and stronger covering constraints; it was introduced by Beeler et al. (2016). Despite a substantial literature on domination in UDGs, DRD has not been studied in UDGs.

2 Preliminaries

2.1 Unit Disk Graphs

A *Unit Disk Graph* (UDG) is a graph $G = (V, E)$ for which each vertex corresponds to a point in the Euclidean plane, and an edge exists between two vertices $u, v \in V$ if and only if the Euclidean distance between the corresponding points satisfies $d(u, v) \leq 1$. Throughout the paper, let $n = |V|$ denote the order of the graph.

A well-known geometric property of UDGs is that at most five mutually independent vertices can lie in the closed neighborhood of any vertex. This packing constraint forms the basis for several of the bounds developed later in this work.

2.2 Double Roman Domination

A *Double Roman Dominating Function* (DRDF) on a graph $G = (V, E)$ is a function

$$f : V \rightarrow \{0, 1, 2, 3\}$$

satisfying the following conditions for every vertex $v \in V$:

- If $f(v) = 0$, then v has either at least two neighbors u with $f(u) = 2$, or at least one neighbor u with $f(u) = 3$.
- If $f(v) = 1$, then v has at least one neighbor u with $f(u) \in \{2, 3\}$.
- If $f(v) \in \{2, 3\}$, no additional requirement is imposed.

The *weight* of f is defined by

$$w(f) = \sum_{v \in V} f(v).$$

The *double Roman domination number* of G is

$$d_R(G) = \min_f w(f),$$

where the minimum is taken over all DRDFs on G .

2.3 Label Sets and Notation

For a DRDF f , the vertex set is naturally partitioned as

$$V_i = \{v \in V : f(v) = i\} \quad \text{for } i \in \{0, 1, 2, 3\}.$$

The set of vertices providing defense is denoted by

$$D = V_2 \cup V_3.$$

The *domination number* of the graph, denoted $\gamma(G)$, is the size of a minimum dominating set in G .

These definitions and notational conventions will be used throughout the remainder of the paper.

3 Research Gap and Problem Statement

3.1 Literature Survey

A comprehensive review was conducted on classical domination and its variants in geometric intersection graphs, particularly focusing on Unit Disk Graphs (UDGs), due to their significant applicability in wireless communication and sensor networks.

The survey covers:

- Classical domination problems: DS, TDS, TRDS, etc.
- Strengthened domination variants: Secure, Restrained, Tuple, Liar domination, etc.
- Roman-style domination variants: Roman domination, Total Roman domination, Roman k -domination, Double Roman domination and Roman $\{2\}$ domination.
- Edge-vertex and vertex-edge domination variants.

For most domination variants in UDGs, the following trend was observed:

- Their decision versions are NP-hard/NP-complete.
- Constant-factor approximation algorithms exist.
- Several PTAS results are available, using local shifting techniques.

3.2 Identified Research Gap

The study reveals that although Roman domination and several strengthened variants have been explored in UDGs, **Double Roman Domination (DRD) has not been studied for Unit Disk Graphs**. Existing works on DRD are limited to general graph classes such as bipartite graphs, and chordal graphs.

Key Observation: While Roman domination variants have received considerable attention, **no literature exists on Double Roman Domination in UDGs**, establishing a strong and clear research gap.

Problem Variant	UDG Complexity	Approximation Results	Status
Roman Domination (RD)	NP-hard	5-approx., PTAS	Well studied
Total Roman Domination (TRD)	NP-complete	Constant-factor approx.	Studied
Double Roman Domination (DRD)	Not known	None	Open in UDGs

Table 1: Research gap analysis for Roman-style domination variants in UDGs.

3.3 Problem Statement

The objective of this research is to study and characterize **Double Roman Domination** in Unit Disk Graphs (UDGs).

Research Objectives

- Establish theoretical bounds for $\gamma_{dR}(G)$ in UDGs.
- Identify structural properties that influence the parameter.
- Investigate the computational complexity of DRD in UDGs.
- Explore approximation or parameterized algorithms, leveraging geometric constraints.

4 Prior Work and Known Results

4.1 Roman Domination in Unit Disk Graphs

The Roman domination problem is known to be **NP-hard** on unit disk graphs (UDGs). Shang and Hu proposed a **linear-time 5-approximation algorithm** based on a maximal independent set strategy, and further developed a **PTAS** (Polynomial-Time Approximation Scheme) using geometric shifting techniques [1]. These results leverage the geometric structure and packing properties of UDGs.

4.2 Double Roman Domination in General Graphs

The Double Roman Domination problem is **NP-hard**, even when restricted to **planar**, **bipartite**, or **chordal** graphs [2]. Known bounds include:

$$2\gamma(G) \leq \gamma_{dR}(G) \leq 3\gamma(G),$$

where $\gamma(G)$ is the domination number [3]. Beeler et al. showed that for any connected graph of order $n \geq 3$,

$$\gamma_{dR}(G) \leq \left\lfloor \frac{6n}{7} \right\rfloor,$$

and this bound is tight for certain graph families [4]. For trees, Nazari-Moghaddam et al. gave a sharper upper bound:

$$\gamma_{dR}(T) \leq \frac{11n - \ell + 4s}{10},$$

where ℓ is the number of leaves and s the number of support vertices [5].

4.3 Algorithms and Approximations

A greedy set cover-based algorithm gives an $O(\log \Delta)$ approximation, where Δ is the maximum degree [6]. Inapproximability results show no polynomial-time algorithm can achieve better than a $(1 - \varepsilon) \ln n$ factor unless $P = NP$. For special cases, such as trees and unicyclic graphs, **linear-time exact algorithms** are known [7].

5 Unconditional Bounds

Theorem 5.1. *For any graph G , the double Roman domination number satisfies*

$$d_R(G) \geq 2\gamma(G).$$

Proof. Let S be a minimum dominating set of G , so $|S| = \gamma(G)$. Consider any DRDF f . For each vertex $s \in S$, either $f(s) \geq 1$ or $f(s) = 0$.

If $f(s) = 0$, then s must be defended by a neighbor of value at least 2, and hence at least one vertex in $N[s]$ has value 2 or 3. In either case, the closed neighborhood of every $s \in S$ contains a vertex assigned value at least 2 under f .

Thus, for each $s \in S$, there exists a vertex $u_s \in N[s]$ with $f(u_s) \geq 2$. Even if multiple vertices of S share the same defender, assigning the defender value 2 contributes at least weight 2 non-negotiably.

Therefore the total weight of f is at least $2|S| = 2\gamma(G)$, and since f was arbitrary, the inequality follows. $\square$

Proposition 5.2. *For any graph G , assigning value 3 to all vertices of a minimum dominating set yields $d_R(G) \leq 3\gamma(G)$.*

Proof. Trivial by construction. $\square$

6 UDG Packing-Derived Bounds

A classical geometric fact about unit disk graphs is the following packing property.

Lemma 6.1 (UDG Packing Lemma). *In any unit disk graph, a vertex can be adjacent to at most five vertices of an independent set.*

This lemma yields a standard lower bound on the domination number of unit disk graphs via maximal independent sets.

Theorem 6.2. *For any unit disk graph $G = (V, E)$ on n vertices,*

$$\gamma(G) \geq \left\lceil \frac{n}{6} \right\rceil.$$

Proof. Let I be a maximal independent set of the unit disk graph $G = (V, E)$. Since I is maximal, every vertex in $V \setminus I$ has at least one neighbor in I , and hence I is a dominating set.

For each vertex $v \in V \setminus I$, choose one neighbor $\phi(v) \in I$. This defines a mapping

$$\phi : V \setminus I \rightarrow I.$$

By Lemma 6.1, a vertex in a unit disk graph can be adjacent to at most five vertices of an independent set. Consequently, each vertex $u \in I$ can be adjacent to at most five vertices $v \in V \setminus I$ with $\phi(v) = u$.

Thus, counting the total number of vertices in $V \setminus I$ via this assignment yields

$$|V \setminus I| \leq 5|I|.$$

Since $|V \setminus I| = n - |I|$, we obtain

$$n - |I| \leq 5|I|,$$

which implies $6|I| \geq n$, and therefore

$$|I| \geq \left\lceil \frac{n}{6} \right\rceil.$$

Finally, as I is a dominating set, $\gamma(G) \leq |I|$, completing the proof. $\square$

As a consequence, we obtain a packing-based lower bound for the double Roman domination number.

Corollary 6.3. *For any unit disk graph G on n vertices,*

$$d_R(G) \geq 2 \left\lceil \frac{n}{6} \right\rceil.$$

Proof. By Theorem 6.2, $\gamma(G) \geq \lceil n/6 \rceil$. Combining this with the unconditional bound $d_R(G) \geq 2\gamma(G)$ yields the result. $\square$

7 Refined lower bound

Let a double Roman dominating function on G be represented by the partition

$$V(G) = V_0 \cup V_1 \cup V_2 \cup V_3,$$

with the usual feasibility conditions. Motivated by the Roman domination literature on unit disk graphs (UDGs), an attempt was made to bound the number of vertices in V_0 that can be “defended” by vertices in $V_2 \cup V_3$ using only geometric packing arguments.

The goal was to establish a refined lower bound of the form

$$d_R(G) \geq 2\gamma(G) + f(n, \gamma(G)),$$

where $f(n, \gamma(G)) > 0$ would arise from the fact that geometric packing limits how many V_0 -vertices can be serviced by a single vertex of value 2 or 3 in a UDG.

While the structural decomposition into V_0 , V_2 and V_3 provides a natural starting point, the detailed counting argument required to turn UDG packing properties into a provable inequality stronger than $2\gamma(G)$ remains difficult. Specifically:

- Bounding the number of V_0 -vertices defended by a vertex in V_2 versus those defended by V_3 is difficult because the way vertices can be adjacent in a unit disk graph is more complicated than what the usual packing rules directly tell us.
- An additional refinement was explored: whether the classical UDG bound that a vertex can have at most five mutually non-adjacent neighbours could be tightened in the double Roman setting. The intuition was that requiring “two-demand” defence might restrict how many V_0 -vertices a single V_2 -vertex could legally defend, potentially reducing the effective packing number below 5. However, this refinement was disproved using a simple geometric construction showing that a V_2 -vertex can still have five “two-demand” neighbours arranged within unit distance without violating UDG constraints. Thus, the classical packing limit of 5 remains tight even under double Roman demand conditions.
- Consequently, no clean UDG-specific inequality of the form

$$d_R(G) \geq 2\gamma(G) + (\text{positive term})$$

has yet been obtained without introducing unverified assumptions.

Therefore, the strongest rigorously established lower bound that applies to all UDGs currently remains the classical general bound

$$2\gamma(G) \leq d_R(G),$$

with the known upper bounds such as $d_R(G) \leq 3\gamma(G)$ and order-based bounds like $d_R(G) \leq 8n/7$ for graphs of minimum degree at least 2 also valid for UDGs as a subclass.

In summary, although several promising geometric pathways were investigated, including a potential refinement of the UDG packing lemma, these did not yield an improvement over the baseline lower bound. The derivation of a genuinely UDG-specific lower bound for $d_R(G)$ remains open.

8 Refined Partition-Based Lower Bound

To obtain sharper geometric lower bounds on the double Roman domination number in unit disk graphs, it is useful to refine the contribution of vertices labeled 0 according to how they are defended. For a DRDF f , the set V_0 is partitioned as

$$V_0 = V_0(2) \cup V_0(3),$$

where

- $V_0(2)$ consists of vertices v with $f(v) = 0$ that have at least two neighbors in V_2 and no neighbors in V_3 ,
- $V_0(3)$ consists of vertices v with $f(v) = 0$ that have at least one neighbor in V_3 .

Denote $a = |V_0(2)|$ and $b = |V_0(3)|$.

Coverage Requirements for the Partition

Coverage requirement for $V_0(2)$. Each vertex $v \in V_0(2)$ requires *two distinct* neighbors in V_2 , contributing a total adjacency demand of

$$2a.$$

By the UDG packing lemma, each vertex in V_2 can be adjacent to at most five vertices of V_0 . Therefore the total adjacency capacity of V_2 is at most $5|V_2|$, and consequently

$$|V_2| \geq \left\lceil \frac{2a}{5} \right\rceil.$$

Coverage requirement for $V_0(3)$. Each vertex $v \in V_0(3)$ requires at least one neighbor in V_3 . Again, the packing lemma implies that each vertex in V_3 can dominate at most five vertices in V_0 , leading to the requirement

$$|V_3| \geq \left\lceil \frac{b}{5} \right\rceil.$$

Theorem 8.1 (Partition bound). *In any UDG G with DRDF f and partition defined above,*

$$d_R(G) \geq 2 \left\lceil \frac{2a}{5} \right\rceil + 3 \left\lceil \frac{b}{5} \right\rceil.$$

Proof. Substituting the coverage bounds

$$|V_2| \geq \left\lceil \frac{2a}{5} \right\rceil, \quad |V_3| \geq \left\lceil \frac{b}{5} \right\rceil,$$

into the DRDF weight expression

$$w(f) = 2|V_2| + 3|V_3|,$$

directly yields the stated inequality. □

Fractional Relaxation

Ignoring ceiling functions gives the relaxed estimate

$$d_R(G) \geq 2 \cdot \frac{2a}{5} + 3 \cdot \frac{b}{5} = \frac{4a + 3b}{5}.$$

Since $|V_0| = a + b$, this implies the simple consequence

$$d_R(G) \geq \frac{3}{5}|V_0|.$$

Although weaker than the integer bound above, this fractional inequality provides an approximation of how the zero-labeled vertices constrain the total weight.

9 Greedy Approximation for Double Roman Domination in Unit Disk Graphs

9.1 Algorithm: Greedy-3-MIS

Algorithm 1 Greedy-3-MIS

Require: A unit disk graph $G = (V, E)$

Ensure: A double Roman dominating function $f : V \rightarrow \{0, 2, 3\}$

- 1: Initialize $f(v) \leftarrow 0$ for all $v \in V$
 - 2: Compute a maximal independent set $I \subseteq V$
 - 3: **for** each $v \in I$ **do**
 - 4: $f(v) \leftarrow 3$
 - 5: **end for**
 - 6: **return** f
-

Lemma 9.1. *Let $G = (V, E)$ be a unit disk graph. The function f returned by GREEDY-3-MIS is a valid double Roman dominating function.*

Proof. By definition, a maximal independent set I is also a dominating set. Hence, for every $u \in V \setminus I$, there exists a neighbor $v \in I$ such that $(u, v) \in E$. Since $f(v) = 3$, all vertices with $f(u) = 0$ have a neighbor assigned 3, satisfying the double Roman domination requirement. As $f(v) \in \{0, 3\}$ and no vertex is assigned 1, the additional constraint for $f(v) = 1$ is trivially avoided. Therefore, f is a valid DRDF. $\square$

9.2 Approximation Guarantee

Theorem 9.2. *The GREEDY-3-MIS algorithm is a constant-factor approximation algorithm for double Roman domination in unit disk graphs. Specifically,*

$$\text{weight}(f) \leq \frac{3c}{2} \cdot \gamma_{\text{dR}}(G),$$

where c is the approximation factor of a maximal independent set for minimum dominating set in UDGs (with $c \leq 5$), and $\gamma_{\text{dR}}(G)$ denotes the double Roman domination number of G .

Proof. Let I be the maximal independent set computed by the algorithm. Since each vertex in I is assigned the value 3, the total weight of the constructed double Roman dominating function is $\text{weight}(f) = 3|I|$.

A classical result of Shang et al. [8] shows that in unit disk graphs, a maximal independent set is a c -approximation to the minimum dominating set, where $c \leq 5$ is derived from geometric packing constraints. Thus,

$$|I| \leq c \cdot \gamma(G),$$

where $\gamma(G)$ denotes the domination number of G .

Beeler et al. [4] established that the double Roman domination number is bounded below by twice the domination number, that is, $\gamma_{dR}(G) \geq 2\gamma(G)$. Combining the inequalities gives

$$\text{weight}(f) = 3|I| \leq 3c \cdot \gamma(G) \leq \frac{3c}{2} \cdot \gamma_{dR}(G),$$

completing the proof. □

10 Complexity of Double Roman Domination in UDGs

Theorem 10.1. *The minimum Double Roman Dominating Set problem is NP-hard on unit disk graphs.*

Proof. The reduction is a polynomial-time reduction from the Minimum Dominating Set problem on grid graphs, which is known to be NP-hard [8]. The reduction closely follows the style of the Roman-domination reduction in [1].

Instance of the source problem. Let $G = (V, E)$ be a grid graph with vertex set

$$V = \{v_1, v_2, \dots, v_n\},$$

and assume each v_i is placed at integer coordinates (x_i, y_i) in the plane, with $(v_i, v_j) \in E$ if and only if the Euclidean distance, $\|(x_i, y_i) - (x_j, y_j)\|_2 = 1$.

Construction of the unit disk graph $G' = (V', E')$. For each original vertex $v_i \in V$ introduce a *duplicate* vertex u_i whose center is placed at $(x_i + \varepsilon, y_i)$ with a small $0 < \varepsilon \ll 1$ (for concreteness one may take $\varepsilon = 1/10$). Define the vertex set

$$V' = \{v_1, u_1, v_2, u_2, \dots, v_n, u_n\}.$$

Put an (undirected) edge between two vertices $a, b \in V'$ iff the Euclidean distance between their centers is at most $1 + \varepsilon$.

By construction each pair (v_i, u_i) are adjacent and, for every i , the closed neighborhoods coincide:

$$N_{G'}[v_i] = N_{G'}[u_i].$$

Also every grid edge $(v_i, v_j) \in E$ gives edges $(v_i, v_j), (v_i, u_j), (u_i, v_j), (u_i, u_j)$ in G' (because distances are at most $1 + \varepsilon$). Thus G' is a unit disk graph.

Intuition of the reduction. To show that G has a dominating set of size at most k if and only if G' has a Double Roman dominating function of weight at most $3k$. The forward direction is immediate (label each chosen dominator with 3).

The following subsection formalizes the steps of the construction.

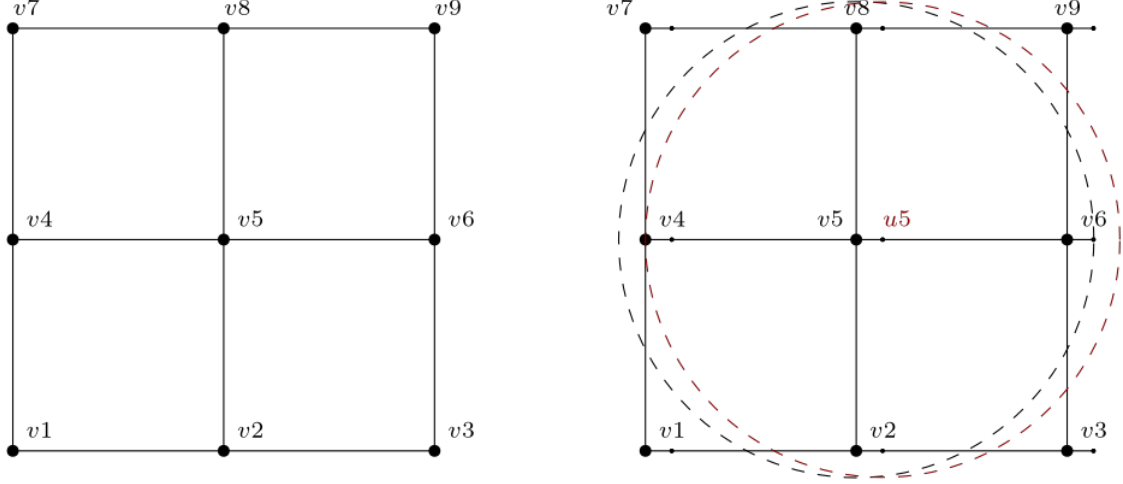


Figure 1: Grid to Unit Disk

(1) Forward direction. Suppose G has a dominating set $D \subseteq V$ with $|D| \leq k$. Define $f : V' \rightarrow \{0, 1, 2, 3\}$ on G' by

$$f(v) = \begin{cases} 3, & \text{if } v \in D, \\ 0, & \text{otherwise,} \end{cases} \quad f(u) = 0 \text{ for all duplicate vertices } u.$$

Every vertex of G' is either in D (thus labeled 3) or adjacent to some $d \in D$ (hence has a neighbor labeled 3), so f is a valid Double Roman dominating function and the weight is $w(f) = 3|D| \leq 3k$. Thus if G admits a dominating set of size k , then G' admits a DRDF of weight $\leq 3k$.

□

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